

Python in Excel

Follow-along worksheet

Your goal today

Use a few Python cells to answer a question from an Excel table.
Then choose one small task to try in your own work.

Name

One Excel task I keep repeating

Before we start

Open follow-along.xlsx. Use this worksheet for today's shorter route; the workbook's opening tab has extra examples. Add a blank sheet named hello-world and move it to the first tab.

Windows: select a cell, then Formulas > Insert Python. Type the code in the formula bar. Ctrl+Enter runs it. Enter adds a line while editing.

1 Your first result

On hello-world, select B2 and insert Python. Type both lines.

```
my_object = "Hello, world!"  
my_object
```

☐ I see Hello, world! (choose Excel Value if needed).

2 Use the name again

In C3, insert Python, type my_object, then run. The same message appears. A name must be created before a later Python cell uses it.

Watching today? Follow the result checks and write down your questions.

Your sales question

Work on the sales sheet. Leave space to the right and below each result.

3 Load the table

Select I1, insert Python, type both lines and run.

```
sales_df = xl("sales[#All]", headers=True)
sales_df.head()
```

Choose Excel Value in the Python output menu to see the rows.
DataFrame means a table in Python. head() shows the first five rows.

The first product I see is

4 Check the numbers

In I10, insert Python, run this line and choose Excel Value.

```
sales_df.describe()
```

Focus on unit_price. Count means how many values; max means largest.

Count

Largest unit price

5 Average unit price by country

In I22, insert Python, run this line and choose Excel Value.

```
sales_df.groupby('country')['unit_price'].mean()
```

Country with the highest average unit price

Its average price

Quick checks after trying: first product Desk; count 200; max 760.69.
Highest average: United Kingdom, about 274.43. This is not total sales or profit.

Two ideas to try next

Watch these demos first. The code is here when you are ready to repeat them.

A chart for each group

On mpg, insert Python in K1. A histogram counts values in each range. Here, each origin gets its own chart with the same scale.

```
mpg_df = xl("mpg[#All]", headers=True)
sns.displot(data=mpg_df, x="mpg", col="origin",
col_order=["usa", "europe", "japan"],
bins=range(5, 56, 5), color="#CF3338")
```

Which group tends to have lower fuel economy in this sample?

A range of possible profits

Add a blank simulation sheet at the end. Insert Python in A1. This example buys 1,100 units for \$30 each and sells each for \$50.

```
np.random.seed(42)
demand = np.maximum(np.random.normal(1000, 200, 10000), 0)
units_sold = np.minimum(demand, 1100)
profit = units_sold * 50 - 1100 * 30
sns.histplot(profit, bins=range(-25000, 25001, 5000),
color="#CF3338")
plt.axvline(0, color="black")
```

Illustration only. Assumes demand centered on 1,000 units, with a standard deviation of 200, bounded at zero. Unsold stock has no resale value. Other costs are excluded. These assumptions drive the results.

Where do the money-losing outcomes sit on the chart?

If you see Image or DataFrame, use the Python output menu. Python in Excel includes the np, sns and plt libraries used here.

Your next step

Use what you tried today on one small, familiar task.

The first result I want to produce in my own workbook

How I will check it against an Excel result I already understand

When I will practice

One question I still have

If you get stuck

No Insert Python: watch along and check access after the session.
#PYTHON!: check spelling, quotes and whether earlier cells have run.
#SPILL!: clear space to the right and below, then run again.
#BUSY!: give the cloud calculation time to finish.

Keep building with the book

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George Mount / Packt

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